

Aravilli Atchuta Ram

Minneapolis, USA

+91-9380653699 | [aravilliatchutaram \[at\] gmail \[dot\] com](mailto:aravilliatchutaram@gmail.com) | [Home Page](#)

EDUCATION

University of Minnesota

PhD in Computer Science

Minneapolis, USA

June 2026 – Present

PES University

Bachelor of Technology in Computer Science & Engineering; GPA 9.23/10.00 $\approx$ 3.95/4.00

Bangalore, India

October 2021 – May 2025

RESEARCH INTERESTS

- TrustworthyAI - AI Safety, Explainable AI
- Deep Learning, Reinforcement Learning

CONFERENCES AND PUBLICATIONS

1. [Aravilli Atchuta Ram](#). **From Vision to Action: Enabling Real-World Agentic VLMs**. *VLM4RWD@NeurIPS2025*
2. [Aravilli Atchuta Ram](#). **Constrained Decoding for Privacy-Preserving LLM Inference**. *ResponsibleFM@NeurIPS 2025*
3. [Aravilli Atchuta Ram](#), Sandarbh Yadav, Yelleti Vivek, and Vadlamani Ravi. **Deep reinforcement learning for financial forecasting in static and streaming cases**. *Journal of Information & Knowledge Management*, 23(06): 2450080, 2024, World Scientific. DOI: 10.1142/S0219649224500801
4. Pranay Gopi, [Aravilli Atchuta Ram](#), Ksnvk Gangadhar, and Vadlamani Ravi. **Classification of Anti-Money Laundering Schemes in Blockchain Networks via Graph Convolution Neural Network based Hybrids**. In *International Conference on Data Management, Analytics & Innovation* (pp. 163-179). Springer Nature Singapore. [Won Best Paper Award]
5. Chilukuri Divyasree, [Aravilli Atchuta Ram](#), and Vadlamani Ravi. **Explainable and Interpretable Isolation Forest for Banking and Finance**. In *International Conference on Data Management, Analytics & Innovation* (pp. 281-304). Springer Nature Singapore. [Won Best Paper Award]
6. Shah, Jay Mintu, Avani Dhagam, Aryan Wadhwa, [Aravilli Atchuta Ram](#), and R. Bharathi. **Recommender Systems for Sector-Specific Stock Analysis**. In *2025 IEEE 14th International Conference on Communication Systems and Network Technologies (CSNT)*, pp. 868–872. IEEE, 2025.

RESEARCH EXPERIENCE

Deep Reinforcement Learning for Financial Forecasting

Supervisor: [Prof. Ravi Vadlamani](#)

IDRBT, Hyderabad, India

May 2023 – August 2023

Description:

- Investigated reinforcement learning approaches (DDPG, PPO, RDPG) for stock market forecasting, benchmarking against baseline ML models (MLP, SVR, GRNN).
- Designed and implemented a Spark Streaming framework with sliding-window forecasting for real-time financial time series analysis.
- Achieved state-of-the-art results: DDPG excelled in static forecasting and GRNN in streaming; findings validated using SMAPE, DS, Theil's U, and Diebold–Mariano tests.

AML Classification in Blockchain via GCN Hybrids

Supervisor: Prof. Ravi Vadlamani

IDRBT, Hyderabad, India

February 2024 – June 2024

Description:

- Tackled the problem of classifying Anti-Money Laundering (AML) schemes in blockchain transaction networks using deep learning.
- Developed hybrid models combining Graph Convolutional Networks (GCNs) with three neural architectures: Probabilistic NN (PNN), Wavelet NN (WNN), and Radial Basis Functional NN (RBFN).
- Evaluated performance on the benchmark Elliptic dataset, achieving superior detection accuracy compared to stand-alone GCN baselines from recent literature.

Explainable and Interpretable Isolation Forest

Supervisor: Prof. Ravi Vadlamani

IDRBT, Hyderabad, India

June 2024 – August 2024

Description:

- Addressed the dual challenge of anomaly detection by enhancing both accuracy and interpretability in high-risk financial domains.
- Proposed a hybrid model integrating Isolation Forest with Decision Trees, combining efficient anomaly identification with rule-based explanations.
- Demonstrated strong detection performance across multiple datasets, while extracted rules provided human-interpretable insights into anomaly causes.

Sector-Specific Stock Recommender Systems

Supervisor: Prof. Bharathi R

PES University, Bengaluru, India

January 2024 – March 2025

Description:

- Designed forecasting models integrating multimodal financial signals (market news, earnings reports, temporal trends, and forecasts) to enable real-time stock price prediction.
- Fine-tuned Llama-3.1-8B-Instruct with QLoRA on a curated financial news corpus to generate sector-specific stock recommendations.
- Proposed a hybrid framework combining price forecasting, sentiment analysis, and performance indicators to derive Buy/Sell decisions, with superior outcomes observed in the consumer cyclical and healthcare sectors.

WORK EXPERIENCE

VISA Inc.	Bangalore, India
Software Engineer	June 2025 – June 2026
• Developed a chatbot for generating SQL and customer queries related to high-volume transaction data. • Tech Stack - Python, MySQL, Docker, Streamlit, Langchain	
Semester Intern	January 2025 – June 2025
• Designed and prototyped RCA Copilot, a multi-agent system with specialized agents for code search, metric analysis, and incident pattern mining. • Collaborated with senior engineers to demonstrate feasibility for automated incident triaging within Visa’s production environment.	
Software Engineering and ML Intern	June 2024 – August 2024
• Developed a scalable embedding-based database comparison tool to validate a custom data replication solution, efficiently handling ~10M-record tables. • Proposed a novel framework leveraging embeddings for large-scale data inconsistency detection. • Recognized with a Technical Innovation Award by the VISA IP Team for impactful innovation.	

AWARDS AND SCHOLARSHIPS

- **Best Paper Awards ICDMAI 2025** — Awarded for 2 papers 2025
- Secured **98.8 percentile** in JEE Mains, among **1.2 million candidates nationwide** 2021
- **Karnataka State Police Hackathon** — 1st place out of 150+ teams 2023
- **HACK'E'LTH Hackathon, GE HealthCare** — 2nd place out of 100+ teams 2023
- **Prof. MRD Scholarship Award (PES University)** — Top 5% among 1000+ students 2024
- **Amazon ML Challenge 2024** — Ranked 65 out of 2000+ teams nationally (Top 5%) 2024

TECHNICAL SKILLS

- Programming Languages:** Python, C, R, Golang, SQL
- ML Frameworks:** PyTorch, Scikit-learn, HuggingFace, NumPy, Pandas
- AI Techniques:** Transfer Learning, Few-Shot Learning, Language Models, Deep Learning, Reinforcement Learning
- Databases:** MongoDB, IBM DB2, MySQL
- Big Data:** Hadoop, Kafka, Zookeeper, PySpark
- Developer Tools:** Git, Jenkins, Docker, Kubernetes, Grafana, Linux